Answers:

Mock Exam 1

Question 1

$U(x, y) = x^{\frac{1}{2}} y^{\frac{1}{4}}$

(a) $U'_x = \frac{1}{2} x^{-\frac{1}{2}} y^{\frac{1}{4}}$

$U'_y = \frac{1}{4} x^{\frac{1}{2}} y^{-\frac{3}{4}}$

(b) $MRS_{xy} = \frac{MU_x}{MU_y}$

Note: the order is important

$= \frac{\frac{1}{2} x^{-\frac{1}{2}} y^{\frac{1}{4}}}{\frac{1}{4} x^{\frac{1}{2}} y^{-\frac{3}{4}}}$

if $MRS_{yx} = \frac{MU_y}{MU_x}$

$= \frac{2y}{x}$

Put this in your cheat sheet!

(c) Relative to y — keep y constant

$\frac{\partial MRS_{xy}}{\partial x} = -\frac{2y}{x^2} < 0$

$\therefore MRS_{xy}$ diminishes

Q2.

(a) $2xy + x^2 \frac{dy}{dx} + 2y \frac{dy}{dx} - 4 = 0$

(b) At $(1, 2)$

$2(1.2) + (1)^2 \frac{dy}{dx} + 2(2) \frac{dy}{dx} - 4 = 0$

$4 + 5\frac{dy}{dx} - 4 = 0$

$\frac{dy}{dx} = 0$

(c) y is neither increasing nor decreasing

since $\frac{dy}{dx} = 0$

Q3.

$f(x,y) = x^2y + 3xy^2$

(a) Step 1: put d

$df = dx^2y + d3xy^2$

Step 2: put constant on the left of d

$df = dx^2y + 3dxy^2$

Step 3: Product Rule

$df = ydx^2 + x^2dy + 3xdy^2 + 3y^2dx$

Step 4: Check if right of d is the simplest form

$df = 2xydx + x^2dy + 6xydy + 3y^2dx$

(b) $\Delta x = 0.1$

$\Delta y = 0.05$

point evaluated is $(2,1)$

$df = 2(2)(1)(0.1) + (2)^2(0.05) + 6(2)(1)(0.05) + 3(1)^2(0.1)$

$= 7(0.1) + 16(0.05)$

$= 1.5$

(c) If productionof x and y increases, how does profit changes

Q4.

(a)

FOC:

$$U'_x = 6x^{-\frac{1}{2}} - 1 = 0$$

$$U'_y = 4y^{-\frac{1}{2}} - 2 = 0$$

(b) Solve

$$\frac{1}{\sqrt{x}} = \frac{1}{6} \qquad \sqrt{x} = 6 \qquad x^* = 36$$

$$\frac{1}{\sqrt{y}} = \frac{1}{2} \qquad \sqrt{y} = 2 \qquad y^* = 4$$

(c) $U'_{x'x} = -3x^{-\frac{3}{2}} < 0$

$U'_{y'y} = -2y^{-\frac{3}{2}} < 0$

$U'_{x'y} = 0$

$U'_{x'x} U'_{y'y} > 0 \qquad (U'_{x'y})^2 = 0^2$

$\therefore U'_{x'x} U'_{y'y} > (U'_{x'y})^2$

$\therefore$ concave

max

Q5.

(a)

$$\pi'_K = 4K^{-\frac{3}{4}}L^{\frac{1}{2}} - 2 = 0$$

$$\pi'_L = 8K^{\frac{1}{4}}L^{-\frac{1}{2}} - 1 = 0$$

(b) Solve

$$\frac{4K^{-\frac{3}{4}}L^{\frac{1}{2}}}{8K^{\frac{1}{4}}L^{-\frac{1}{2}}} = \frac{2}{1}$$

$$\frac{L}{2K} = 2$$

$$L = 4K$$

$$4K^{-\frac{3}{4}}(4K)^{\frac{1}{2}} = 2$$

$$4K^{-\frac{3}{4}}4^{\frac{1}{2}}K^{\frac{1}{2}} = 2$$

$$8K^{-\frac{1}{4}} = 2$$

$$K^{-\frac{1}{4}} = \frac{1}{4}$$

$$K^* = \left(\frac{1}{4}\right)^{-4}$$

$$= 256$$

$$L^* = 1024$$

(c) $\pi'_{KK} = -3K^{-\frac{7}{4}}L^{\frac{1}{2}} = -\frac{3}{512} < 0$

$\pi'_{LL} = -4K^{\frac{1}{4}}L^{-\frac{3}{2}} = -\frac{1}{2048} < 0$

$\pi'_{KL} = 2K^{-\frac{3}{4}}L^{-\frac{1}{2}} = \frac{1}{1024}$

$(\pi'_{KL})^2 = \left(\frac{1}{1024}\right)^2$

$(\pi'_{KK}\pi'_{LL}) = \frac{3}{(512)(2048)} > \frac{1}{(1024)(1024)}$

∴ Concave, max

Mock Exam 2

Q1

(a)

$U'_x = 2(x+1)(y+2)^3$

$U'_y = (x+1)^2 \cdot 3(y+2)^2$

(b) $$MRS_{xy} = \frac{2\cancel{(x+1)}(y+2)^{\cancel{3}}}{(x+1)^{\cancel{2}} \cdot 3\cancel{(y+2)^2}}$$

$$= \frac{2(y+2)}{3(x+1)}$$

(c) $$\frac{\partial MRS_{xy}}{\partial x} = \frac{2(y+2)}{3}(-1)(x+1)^{-2} < 0$$

$$\frac{\partial MRS_{xy}}{\partial y} = \frac{2}{3(x+1)} > 0$$

$\therefore$ MRS changes with x and y, willingness to sub will change

Q2.

(a) $2x + \quad y + x\frac{dy}{dx} \quad + 2y\frac{dy}{dx} = 0$

(b) At $(2,2)$

$4 \quad + 2 \quad 2\frac{dy}{dx} + \quad 4\frac{dy}{dx} = 0$

$6\frac{dy}{dx} = -6$

$\frac{dy}{dx} = -1$

(c) $y = -x + c$

$2 = -2 + c$

$c = 4$

$\therefore \; y = -x + 4$

Q3.

(a) $dz = dx^3y^2 + 2dxy$

$dz = x^3dy^2 + y^2dx^3 + 2(ydx + xdy)$

$dz = 2yx^3dy + 3x^2y^2dx + 2ydx + 2xdy$

(b) $\Delta x = 0.02 \quad \Delta y = -0.03$

evaluated at $(1,2)$

$dz = 2(2)(1)^3(-0.03) + 3(1)^2(2)^2(0.02)$

$\quad + 2(2)(0.02) + 2(1)(-0.03)$

$= 16(0.02) + 6(-0.03)$

$= 0.14$

(c) same as Mock Exam 1

Q4.

a. FOC

$C'_x = x + y + 2 = 0$

$C'_y = x + 2y + 1 = 0$

(b)

$y = -2 - x$

$x + 2(-2 - x) + 1 = 0$

$x - 4 - 2x + 1 = 0$

$-x - 3 = 0$

$x^* = -3$

$y^* = 1$

(c) $C'_{x'x} = 1 > 0$

$C'_{y'y} = 2 > 0$

$C'_{x'y} = 1$

$C'_{x'x} C'_{y'y} = 1 \times 2 = 2$

$(C'_{x'y})^2 = 1^2 = 1$

$C'_{x'x} C'_{y'y} > (C'_{x'y})^2$

$\therefore$ Convex minimum

Q5.

(a) FOC

$\pi'_K = \frac{18}{4} K^{-\frac{3}{4}} L^{\frac{1}{2}} - 1 = 0$

$\pi'_L = \frac{18}{2} K^{\frac{1}{4}} L^{-\frac{1}{2}} - 2 = 0$

(b)

$$\frac{\frac{18}{4} K^{-\frac{3}{4}} L^{\frac{1}{2}}}{\frac{18}{2} K^{\frac{1}{4}} L^{-\frac{1}{2}}} = \frac{1}{2}$$

$$\frac{L}{2K} = \frac{1}{2}$$

$$L = K$$

$$\frac{18}{4} K^{-\frac{3}{4}} K^{\frac{1}{2}} - 1 = 0$$

$$\frac{18}{4} K^{-\frac{1}{4}} = 1$$

$$K^{-\frac{1}{4}} = \frac{4}{18}$$

$$K = \left(\frac{4}{18}\right)^{-4}$$

$$= \frac{6561}{16}$$

$$L = \frac{6561}{16}$$

(c) $\pi'_{K'K} = -\frac{3}{4} \cdot \frac{18}{4} K^{-\frac{7}{4}} L^{\frac{1}{2}} = -\frac{128}{2187} < 0$

$\pi'_{L'L} = -\frac{1}{2} \cdot \frac{18}{2} K^{\frac{1}{4}} L^{-\frac{3}{2}} = -\frac{512}{6561} < 0$

$\pi'_{K'L} = \frac{18}{4} \cdot \frac{1}{2} K^{-\frac{3}{4}} L^{-\frac{1}{2}} = \frac{256}{6561} \approx (0.0390)^2$

$(\pi'_{K'L})^2 = \left(\frac{256}{6561}\right)^2$

$\pi'_{K'K}\pi'_{L'L} \approx 0.00362$

$\pi'_{K'K}\pi'_{L'L} - (\pi'_{K'L})^2 = 0.00362 > 0$

$\therefore$ Convex

Max